Candidate Seat no…….

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech (IT) Examination

Nov 2019

IT444 Internet of Things

Date: 11.11.2019, Monday Time: 01.30 p.m. To 04.30 p.m. Maximum Marks: 70

Instructions:

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 Do as directed. [07]

(a) Full form of MQTT is _____. [01]

(i) Multiple Queuing Telegram Transport
(ii) Multiple Queuing Telemetry Transport
(iii) Message Queuing Telegram Transport
(iv) Message Queuing Telemetry Transport

(b) **State True/False with appropriate justification.** [02]

(i) AMQP protocol is used for real time communication.
(ii) Email Notification is example of push technology.

(c) **Differentiate** [04]

(i) Arduino UNO v/s ESP32
(ii) M2M v/s IoT

Q-2.a List various communication model. Explain Publish - Subscribe Model in detail. **[04]**

Q-2.b Answer any TWO questions. [10]

(i) Draw and explain IoTWF standardized architecture in detail.

(ii) Compare IoT levels in detail with example.

(iii) Discuss various fundamental characteristics of IoT in detail. List various diversified application of IoT based on industry perspective.

Q-3 Answer any TWO questions. [14]

(i) Discuss RPL control messages using sequence diagram. Elaborate working of RPL protocol using flowchart and example scenario.

(ii) Prepare IoT system design methodology for Home Intrusion Detection System.

(iii) Define Internet of Things. Explain four pillars of IoT in detail.

SECTION – II

Q - 4 Do as directed. **[07]**

(a) What is the size of IPv6 address? [01]

(i) 32-bit (iii)128-bit

(ii) 64-bit (iv)256-bit

(b) **State True/False with appropriate justification.** [02]

(i) RPL supports P2P, MP2P and P2MP traffic.

(ii) IIoT means Instrumentation Internet of Things.

(c) **Differentiate** [04]

(i) Raspberry Pi3 v/s Raspberry Pi 4

(ii) REST v/s Websocket communication API

Q–5.a Elaborate in detail various web services provided by AWS for IoT. **[04]**

Q–5.b Answer any TWO questions. **[10]**

(i) With the help of suitable example explain in detail role of sensors and actuators while designing IoT applications.

(ii) What is RFID? Explain the challenges and issues facing the deployment of RFID Technology.

(iii) What are the types of security attacks? List IoT security problems. Explain in detail security problem caused on open ports.

Q–6 Answer any TWO questions. **[14]**

(i) Enlist wired and wireless connectivity for Internet of Things. Differentiate Wi-Fi v/s Zigbee v/s Bluetooth.

(ii) Explain IoT vulnerabilities along with obstacles and its solutions.

(iii) Discuss IoT Protocol Stack in detail with neat diagram.

***